Md Kamruzzaman

AI & Python Backend Engineer | FastAPI, REST APIs, PostgreSQL | RAG, LLM, MLOps

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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

B.Tech Computer Science Graduate (Machine Learning specialization, 2025) with hands-on experience building and deploying FastAPI REST APIs, PostgreSQL schemas, and RAG pipelines using LangChain and ChromaDB. Portfolio includes live, working demos covering JWT authentication, model serving, Docker, MLflow, and CI/CD. Seeking a Python Backend or AI/ML role to contribute to clean APIs, reliable model serving, and monitored AI systems. Open to remote roles globally; available to overlap with global time zones.

TECHNICAL SKILLS

Backend & APIs: Python, FastAPI, REST APIs, SQLAlchemy, Alembic, JWT Authentication, Pydantic
Generative AI: LangChain, RAG Pipelines, ChromaDB, Google Gemini API, Prompt Engineering
Machine Learning: scikit-learn, Logistic Regression, Gaussian Naive Bayes, TF-IDF, NLP, classification
MLOps & Monitoring: MLflow, Prometheus, Grafana, Evidently (Drift Detection), Model Serving
DevOps & Databases: Docker, Kubernetes, CI/CD (GitHub Actions), Git, PostgreSQL, SQL

PRACTICAL EXPERIENCE

Backend & AI Developer (Independent Project)

Expense Tracker API — Backend Engineering & Deployment March 2026 – October 2026
Code | Live API

- Engineered a secure REST API with **8 endpoints** using FastAPI, implementing JWT authentication and bcrypt hashing to enforce user-scoped data isolation.
- Designed a normalized PostgreSQL schema with SQLAlchemy and Alembic, utilizing Pydantic for strict data validation, pagination, and error handling.
- Achieved **91% test coverage** across authentication and CRUD flows with **11 automated pytest tests**, executing via GitHub Actions CI/CD on every commit.
- Optimized database queries to achieve **∼10ms local latency** and **sub-300ms live cloud latency** (deployed on Render/Neon).

ML Expense Platform — MLOps & Model Serving September 2026 – October 2026
Code | Live API

- Architected an end-to-end ML serving pipeline using FastAPI, scikit-learn, and MLflow to classify free-text expense descriptions.
- Trained a TF-IDF and Logistic Regression model on a 37-row synthetic dataset, tracking and comparing **4 hyperparameter tuning runs** via MLflow.
- Achieved **∼5ms inference latency** and handled **500+ requests/second** by loading the model into RAM at API startup with Pydantic validation.
- Containerized the application with Docker and authored Kubernetes manifests; instrumented Prometheus metrics and Grafana dashboards for monitoring and Evidently-based data drift.

Docu-RAG API — Generative AI & LLM Integration October 2026 – Present
Code

- Developed a Retrieval-Augmented Generation (RAG) API using FastAPI, LangChain, and ChromaDB to enable context-aware question answering over PDF documents.
- Engineered a document ingestion pipeline using `RecursiveCharacterTextSplitter` (500-character chunks, 50-character overlap) to generate and store vector embeddings.
- Integrated the Google Gemini API for semantic search and answer generation, achieving a **2–4 second end-to-end latency** by optimizing the retrieval flow.
- Containerized the API using Docker and conducted manual testing via Swagger UI to ensure factual grounding; planned integration of **RAGAS** for automated evaluation.

EDUCATION

Lovely Professional University, Punjab, India 2025
B.Tech in Computer Science and Engineering, Machine Learning Specialization

CERTIFICATIONS

Natural Language Processing Specialization (DeepLearning.AI) | Meta Back-End Developer (Coursera) | Python Data Structures (University of Michigan)